

Supplementary Material for “What Do Wrist-Worn Stress Detectors Detect? An Untrained Arousal Index Nearly Matches Them Across Five Empatica E4 Datasets”

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TABLE SI
TEST-SIDE ERROR BUDGET: BALANCED-ACCURACY GAIN OF EACH FILTER, SHAPLEY-AVERAGED OVER 24 ORDERS (HR/HRV/EDA)

Dataset, model	All	F1	F2	F3	F4	Resid.
PhysioNet, within	0.776	0.003	0.007	0.020	0.020	0.175
PhysioNet, external	0.746	0.015	0.006	0.036	0.008	0.190
Stress-Predict, within	0.710	0.019	0.011	0.018	0.012	0.229
Stress-Predict, external	0.653	0.015	−0.015	0.012	0.005	0.329
WESAD, external	0.887	0	0.045	0	0.002	0.066

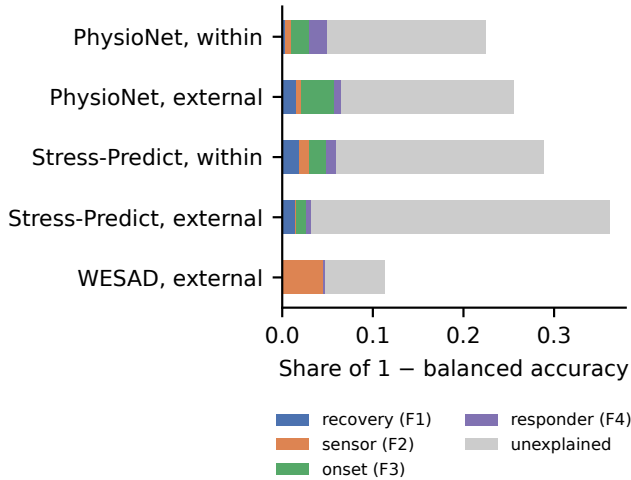


Fig. S1. Error budget. Each bar is 1–BA of the out-of-sample predictions; coloured segments are the Shapley-averaged gains of the four test-side filters, grey is the residual after all four. Negative gains (Stress-Predict external, F2) are drawn as zero.

Section, table and figure numbers refer to this supplement. All values come from committed tables in <https://github.com/ALVAROI12/smartwatch-stress-detection>.

S-I. ERROR BUDGET AND THE CHEST-ECG BOUND

Table SI and Fig. S1 test four explanations for the low ceilings on PhysioNet and Stress-Predict as test-side filters on LOSO and external predictions: F1 drops rest windows within 5 min of a stressor offset (recovery contamination); F2 keeps windows with cardiac coverage of at least 0.5

(sensor loss); F3 drops the first 60 s of each stressor (onset latency); F4 keeps only subjects whose own stressor-minus-baseline heart rate or tonic EDA exceeds 0.5 baseline SD (non-response; label-using, diagnostic only). Together they buy 0.05–0.06 BA on PhysioNet and Stress-Predict (0.02 for the external model on Stress-Predict). With subject-bootstrap CIs (1000 draws; `rev8_budget_bootstrap_ci.csv`), F1 and F3 exclude zero on Stress-Predict, F3 on external PhysioNet and F2 on WESAD; no single filter’s upper bound exceeds 0.065 on PhysioNet or Stress-Predict. Without the label-using F4, the three label-free filters buy at most 0.056 (upper bound 0.09), and F4 adds at most 0.022 on top. The residual error of 0.17–0.23 (0.33 external on Stress-Predict; lower CI bounds 0.11–0.28) is explained by none of them. The responder filter is the one term sensitive to its definition (`partB_responder_sensitivity.csv`). At 0.5 or 1.0 baseline SD on either channel the two rules select the same subjects (32 of 35 PhysioNet, 32 of 34 Stress-Predict) and F4 gains 0.01–0.02. Requiring 1.0 SD on both channels keeps 17 of 35 PhysioNet subjects and raises within-dataset BA by 0.094 [0.05, 0.15] (external 0.065 [0.01, 0.13]); keeping the top half of subjects by pooled response gives 0.062 [0.01, 0.13]. On Stress-Predict the strict rules keep 12–17 of 34 subjects and gain 0.03 within (CI includes zero) and −0.02 to +0.03 externally. Under the strict rules non-response is the largest single term on PhysioNet, about half of the four-filter total, yet the residual stays at 0.10–0.14 on PhysioNet and 0.21–0.35 on Stress-Predict. Onset latency and recovery are real but small: within-dataset Stress-Predict recall is 0.47 in the first minute of a stressor against 0.77 after 3 min, PhysioNet 0.63 in the first minute against 0.77 in the third, and rest false alarms peak in the first minute after offset (PhysioNet within 0.26 against 0.06–0.09 later); too few windows sit there to move BA.

On WESAD the only sizeable term is the sensor (F2, +0.045 external). Replacing wrist cardiac features with chest-ECG features on the same windows raises BA from 0.916 to 0.956 (AUROC 0.977 to 0.993), and chest ECG alone reaches 0.918, equal to the whole wrist pipeline; the gain concentrates in low-coverage windows, where wrist heart rate is absent (0.886 to 0.962). EDA alone gives 0.891. Motion-robust wrist heart rate is therefore worth at most about +0.04 on WESAD, and about 0.04 of error remains with chest-grade cardiac data.

TABLE SII
LEAKAGE-CONTROLLED NEGATIVE RESULTS (BALANCED ACCURACY,
HR/HRV/EDA UNLESS STATED)

Question	Result
Unsupervised domain adaptation (correlation alignment, maximum mean discrepancy, domain-adversarial and subject-adversarial networks), WESAD $\leftrightarrow$ PhysioNet, physiology	0.52–0.70 vs source-only 0.54–0.70; none reliably better
Few-shot on unseen dataset, first-in-time windows, $k = 1$ –5 per class	–0.04 to +0.04 refit gain; none significant after Holm
Few-shot, random windows	only significant gains (WESAD $k = 5$, +0.03); not deployable
Per-user threshold from calibration windows	hurts PhysioNet (-0.08 , $p_{\text{holm}} < 0.05$ at $k = 3$)
Self-rated stress rise vs window recall (PhysioNet 54 units, WESAD 15)	PhysioNet $\rho = -0.08$ [$-0.42, 0.21$] within, 0.02 [$-0.30, 0.31$] external; WESAD -0.49 [$-0.87, 0.17$] within, -0.60 [$-0.84, -0.08$] external (15 units); mild tasks not missed more
Detectability as a subject trait: recall across stressors	Stress-Predict $\rho = -0.14$ [$-0.51, 0.24$]; PhysioNet 0.44 [$-0.04, 0.78$]
Recall vs baseline HR, RMSSD, EDA, PPG coverage	$ \rho \leq 0.35$ with CIs including 0, except Stress-Predict external recall vs baseline EDA ($\rho = 0.48$ [$0.18, 0.70$])
Worst 20% of subjects' share of misses	37–61%
Exercise absent from training (LODO with exercise as negatives)	69.5% of exercise windows called stress by the other-4 model; 4–6% when PhysioNet's own subjects are in training (8–10% under the panel's added-negative configuration, main-text Table III)

S-II. LEAKAGE-CONTROLLED NEGATIVE RESULTS

Table SII collects results that close directions the literature has proposed. Unsupervised domain adaptation on top of per-subject normalisation does not beat source-only training in either direction between WESAD and PhysioNet, consistent with Kwon et al. [2]. Few-shot personalisation of a model trained on the other four datasets, with each user's first k windows per class as calibration and a 30-s gap before any test window, adds at most 0.02–0.04 BA and nothing significant; drawing calibration windows at random from the recording produces the only significant gains, in line with Tervonen et al. [3] and Akkaya [1]. A per-user threshold hurts. Self-rated stress intensity does not predict which windows are missed: in PhysioNet, Stroop (window-weighted mean rise 0.7 of 10) is detected as well as arithmetic (2.5); WESAD's negative correlation rests on the two subjects with the largest rise. The data give no support to detectability as a stable subject trait: recall is not correlated across stressors, is not predicted by baseline physiology or PPG coverage (one exception, Stress-Predict external recall against baseline EDA), and misses are only moderately concentrated; PhysioNet's within and external models do agree on who is hard ($\rho = 0.64$ [$0.37, 0.82$]).

REFERENCES

- [1] A. Akkaya, "Calibration, not architecture, limits cross-subject wearable stress detection: A multi-dataset, decision-focused evaluation with label-

- efficient recalibration," *BMC Med. Inform. Decis. Mak.*, 2026, doi: 10.1186/s12911-026-03842-1 (abstract only).
 [2] N. Kwon, S. Yoon, J. Hur, and H. S. Kang, "When wellness stress detection enters healthcare workflows: An iso-heart-rate, cross-corpus evaluation of wrist wearables," *Healthcare*, vol. 14, no. 15, Art. no. 2434, 2026, doi: 10.3390/healthcare14152434.
 [3] J. Tervonen, E. Amparore, M. Botta, J. Närväinen, K. Pettersson, and J. Mäntyjärvi, "Data-constrained personalization of cognitive state detection with feature-based and foundation models," *User Model. User-Adapt. Interact.*, 2026, doi: 10.1007/s11257-026-09444-w.